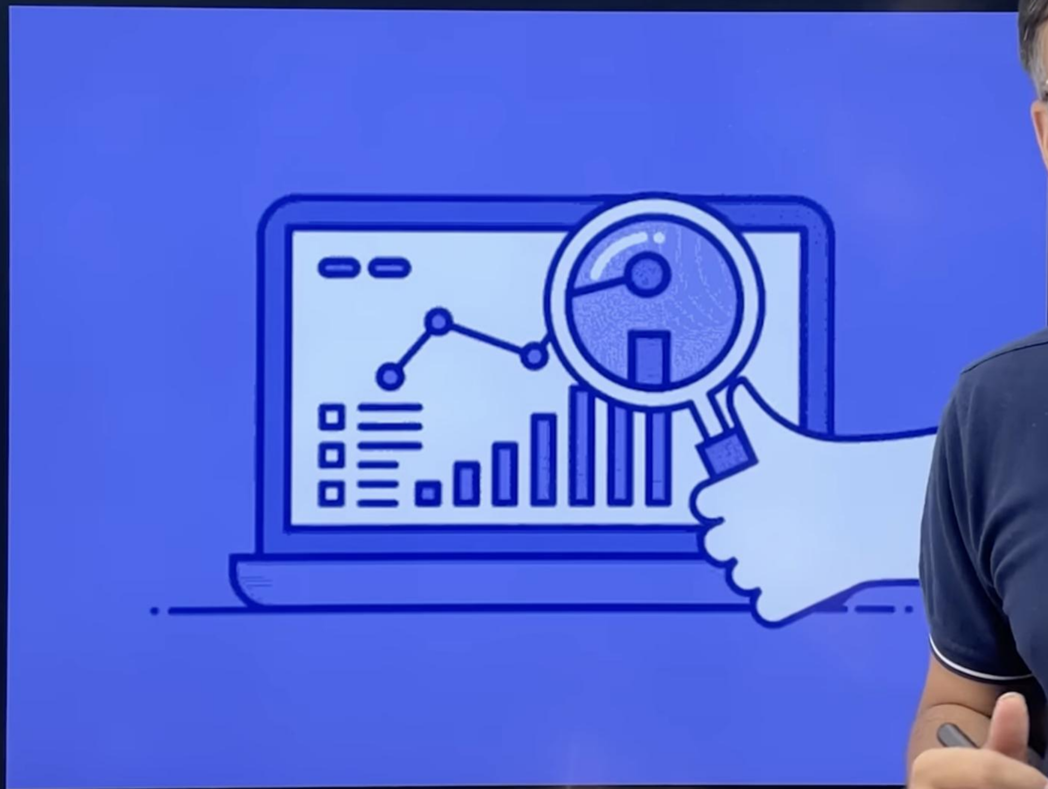
What is Data ?

- Data are characteristics or attributes, often numerical, collected through observation and can be qualitative(descriptive) or quantitative(numerical). Any facts and figures about an entity is called as Data.
- Data serves a crucial role in various sectors by facilitating analysis, supporting decision-making processes, and being fundamental to research activities.

What is Information ?

- Data becomes information when analyzed and placed in context, providing a basis for understanding, decision-making, and further analysis. Processed Da lled information.

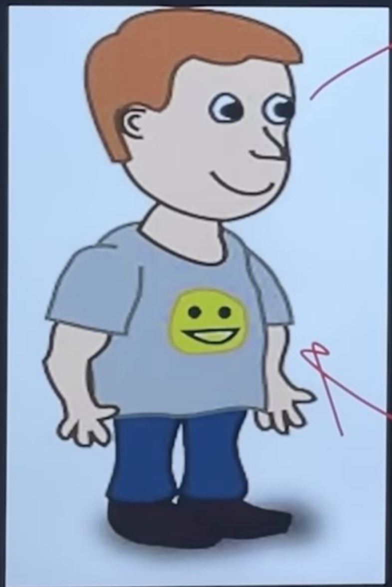


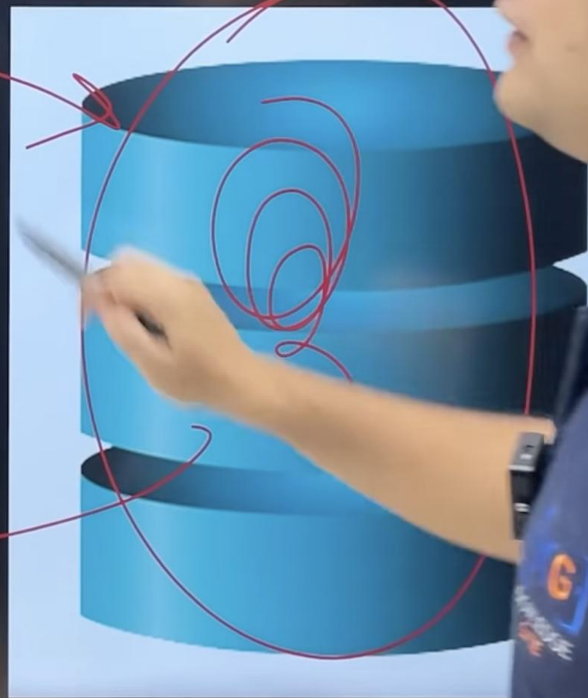
What is Data Base?

- A database is a structured collection of data, facilitating easy access, management, and updates., generally stored and accessed electronically from a computer system.

What is Data Base Management System?

- A DBMS is software facilitating efficient data storage, retrieval, and management in databases.
- Ensures data safety and integrity, while offering accessibility and conc
- Supports functions like data querying, reporting, and analytics for in making.





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Chapter-1 (Basics)

Database System vs File System

Aspect	File System	Database Management System
Data Access	Slower data retrieval due to unstructured querying capabilities.	Structured querying capabilities allow for quicker data access.
Data Isolation	Challenges in correlating data across separate files leading to data isolation.	Facilitates data integration, reducing data isolation issues.
Data Integrity	Risk of inadvertent data alterations or deletions creating integrity problems.	Features to prevent unauthorized data alterations, maintaining integrity.
Atomicity Problem	Potential for data inconsistency due to incomplete operations, leading to atomicity problems.	Supports transaction properties like atomicity, ensuring operations are either completed fully or not at all.
Concurrent Access Anomalies	Conflicts and inconsistencies from simultaneous data access/modifications, causing concurrent access anomalies.	Advanced concurrency controls to manage multiple users accessing the database simultaneously, reducing anomalies.

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9:55 / 5:33:30 • (Chapter-1: Basics)- Data & information, Database System vs File System, Views of D... ›

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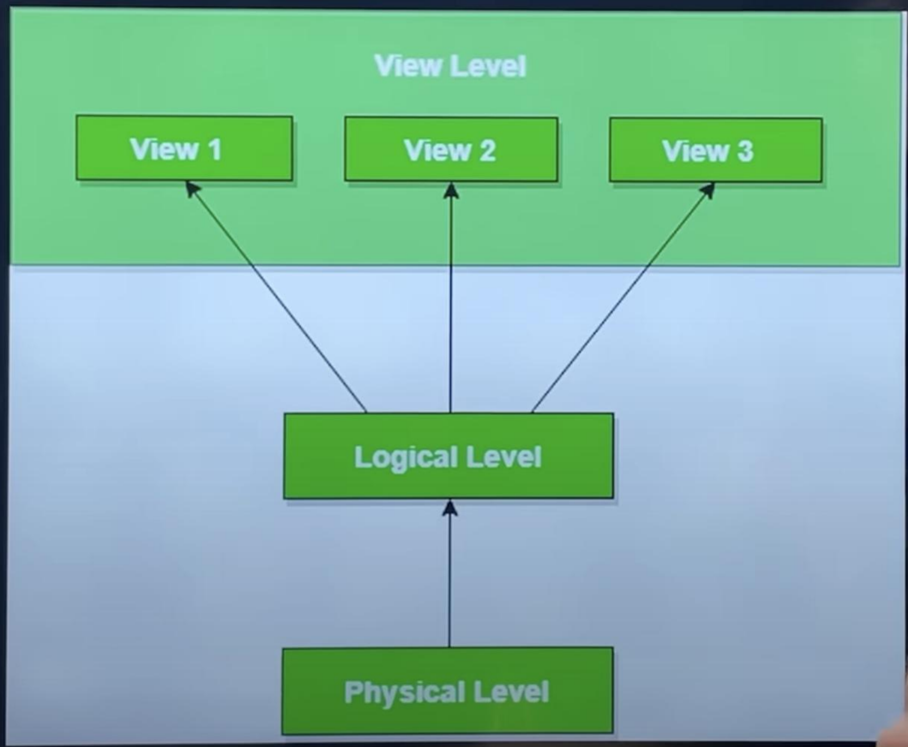
Chapter-1 (Basics)

Views of Data Base

View of Data Base (Data Abstraction)

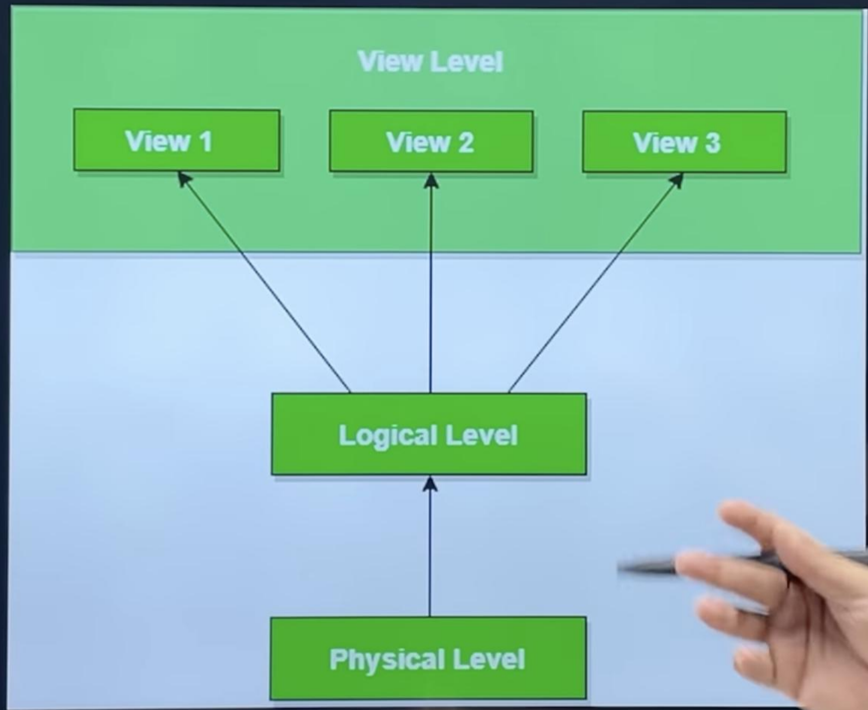
- **Physical Level:** The internal schema details data storage and access on hardware, featuring the lowest level of data abstraction with complex structures, predominantly managed by the database administrator.



11:59 / 5:33:30 • (Chapter-1: Basics)- Data & information, Database System vs File System, Views of ...

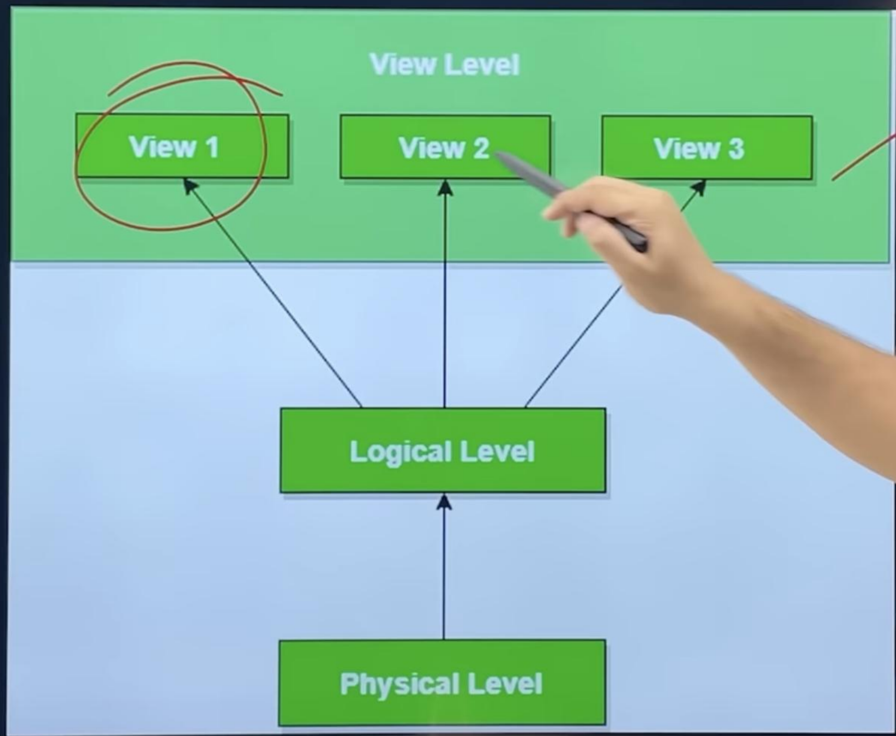
View of Data Base

- **Logical Level/ Conceptual Level**: Above the physical level, this level showcases data as entity sets and their relationships, detailing the types and connections between stored data in the database.



View of Data Base

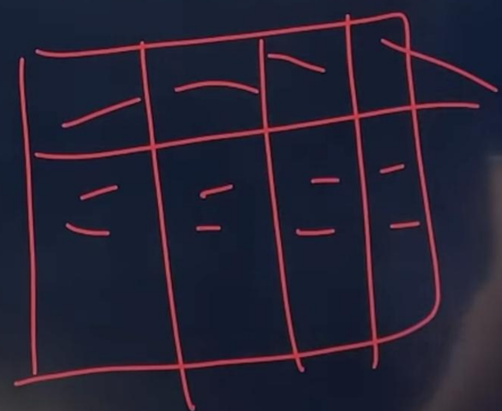
- **View Level**: This is the pinnacle of data abstraction, displaying only a portion of the entire database focusing on user-interest areas. It can represent multipl views of the same data, allowing users to access information through vario applications from the database.

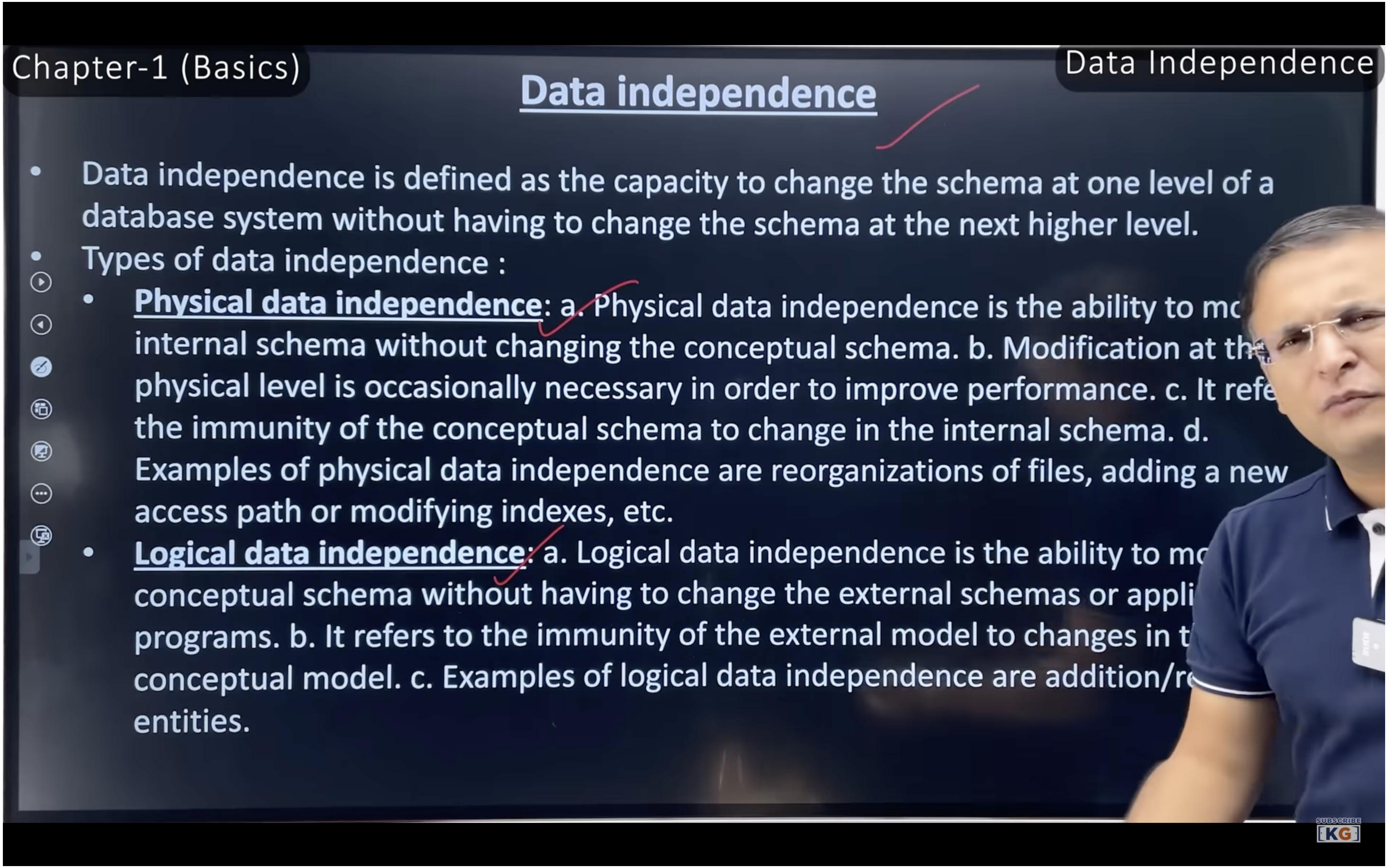


Instance and Schemas

- **Instance of the Database**:
  - The collection of information stored in the database at a specific moment is known as an instance of the database. It is a snapshot of the database that contains live data moment, showing the current state of all records and transactions.

- **Database Schema**:
  - The database schema refers to the overall design of the database, illustrating the lo structure and organization of data. It defines how data is organized and how relationsh between data are handled, essentially serving as the blueprint for how the database i constructed.




Chapter-1 (Basics)
Data Independence
Data independence
Data independence is defined as the capacity to change the schema at one level of a database system without having to change the schema at the next higher level.
Types of data independence :
Physical data independence: a. Physical data independence is the ability to mo
internal schema without changing the conceptual schema. b. Modification at th
physical level is occasionally necessary in order to improve performance. c. It refe
the immunity of the conceptual schema to change in the internal schema. d.
Examples of physical data independence are reorganizations of files, adding a new
access path or modifying indexes, etc.
Logical data independence: a. Logical data independence is the ability to mo
conceptual schema without having to change the external schemas or appli
programs. b. It refers to the immunity of the external model to changes in t
conceptual model. c. Examples of logical data independence are addition/re
entities.
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Aspect	OLAP (Online Analytical Processing)	OLTP (Online Transaction Processing)
Primary Function	Designed for complex data analysis and reporting.	Handles daily transactional data processing.
Database Design	Star or snowflake schema, optimizing for read operations.	Normalized schema, optimizing for write operations.
Query Complexity	Complex queries involving aggregations and computations across multiple dimensions.	Simple and standard queries focusing on CRUD operations (Create, Read, Update, Delete).
Data Volume	Deals with large volumes of data for historical analysis.	Processes a high number of small transactions.
Response Time	Slower response time due to complex queries.	Fast response time to support high transaction rates.

Types of Data Base

- **Commercial Database**: Predominantly used in the business sector to handle large volumes of transactions and customer data. A CRM system like Salesforce which handles large volumes of customer data and transactions.
- **Multimedia Database**: Stores data types such as images, audio, and video files, facilitating the management and retrieval of multimedia content. A digital asset management system like Adobe Experience Manager that facilitates the storage and retrieval of multimedia content.
- **Deductive Database**: Utilizes logic programming to derive information from data stored in a database, allowing for more complex and analytical queries. A database using Datalog (a query language) which allows for complex logical queries and information derivation.

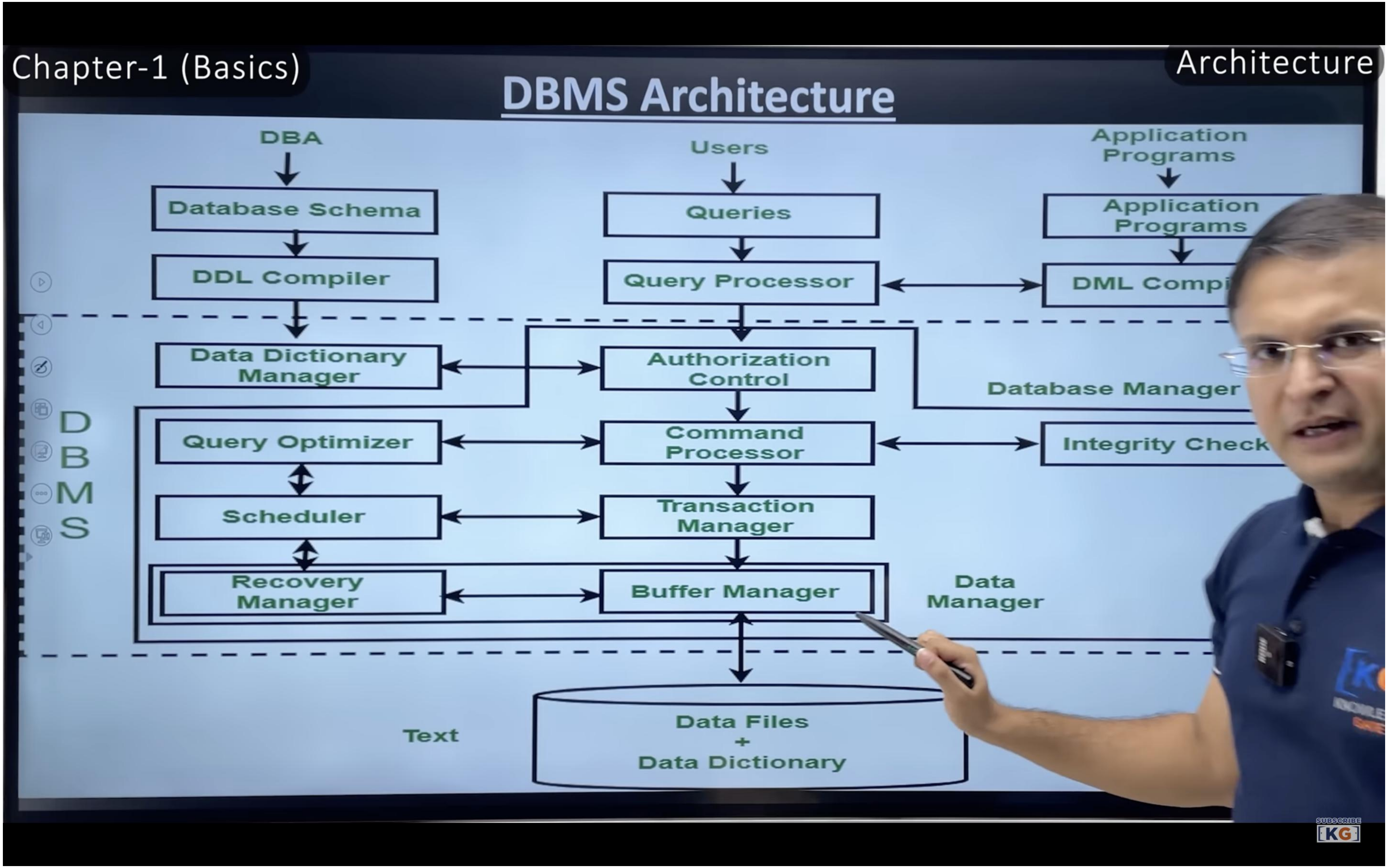
Temporal Database: Keeps track of changing data over time, allowing for queries concerning time-based data. A historical trading database in the financial sector which keeps track of stock prices over time.

Geological Information System (GIS): Stores, organizes, and analyzes geographical data, aiding in spatial analysis and mapping projects. A system like ArcGIS which enables the storage, analysis, and visualization of geographical data.

DBA(Database Administrator)

Database administrators hold authority over data and the programs facilitating data access. Their roles/functions are:

- **Schema Definition**: DBA outlines the original database schema. Achieved through writing definitions translated to permanent labels in the data dictionary by the DDL compiler.
- **Storage Structure and Access Method Definition**: Responsible for forming appropriate storage structures and access methods. Achieved through writing definitions translated by the data storage and definition language compiler.
- **Schema and Physical Organization Modification**: Involves altering the database schema or physical storage organization. Changes are implemented by writing definitions that modify the relevant internal system tables.
- **Granting of Authorization for Data Access**: DBA grants varied types of data access authorization to different database users.
- **Integrity Constraint Specification**: DBA implements and maintains integrity constraints to ensure data accuracy and consistency.


Chapter-1 (Basics)
Architecture
DBMS Architecture
DBA
Users
Application Programs
Database Schema
Queries
Application Programs
DDL Compiler
Query Processor
DML Compi
Data Dictionary Manager
Authorization Control
Database Manager
D B M S
Query Optimizer
Command Processor
Integrity Check
Scheduler
Transaction Manager
Recovery Manager
Buffer Manager
Data Manager
Text
Data Files + Data Dictionary

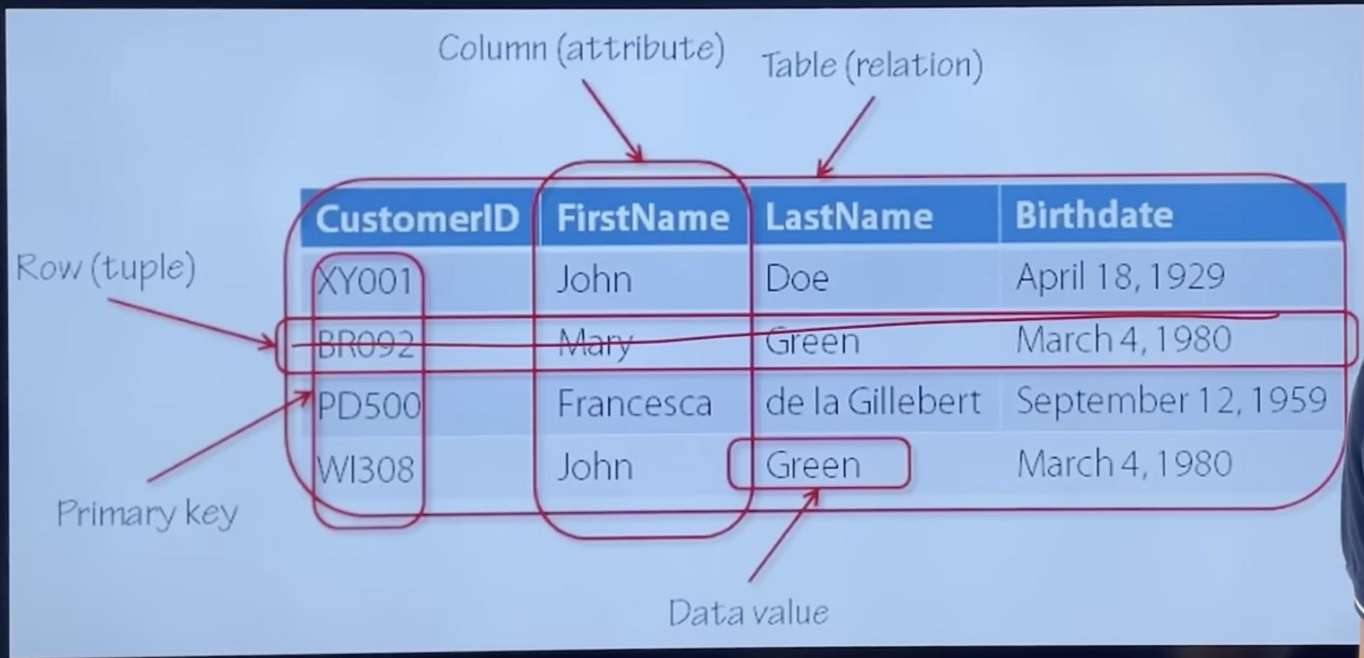
ENTITY

- An entity is a thing or an object in the real world that is distinguishable from other object based on the values of the attributes it possesses.

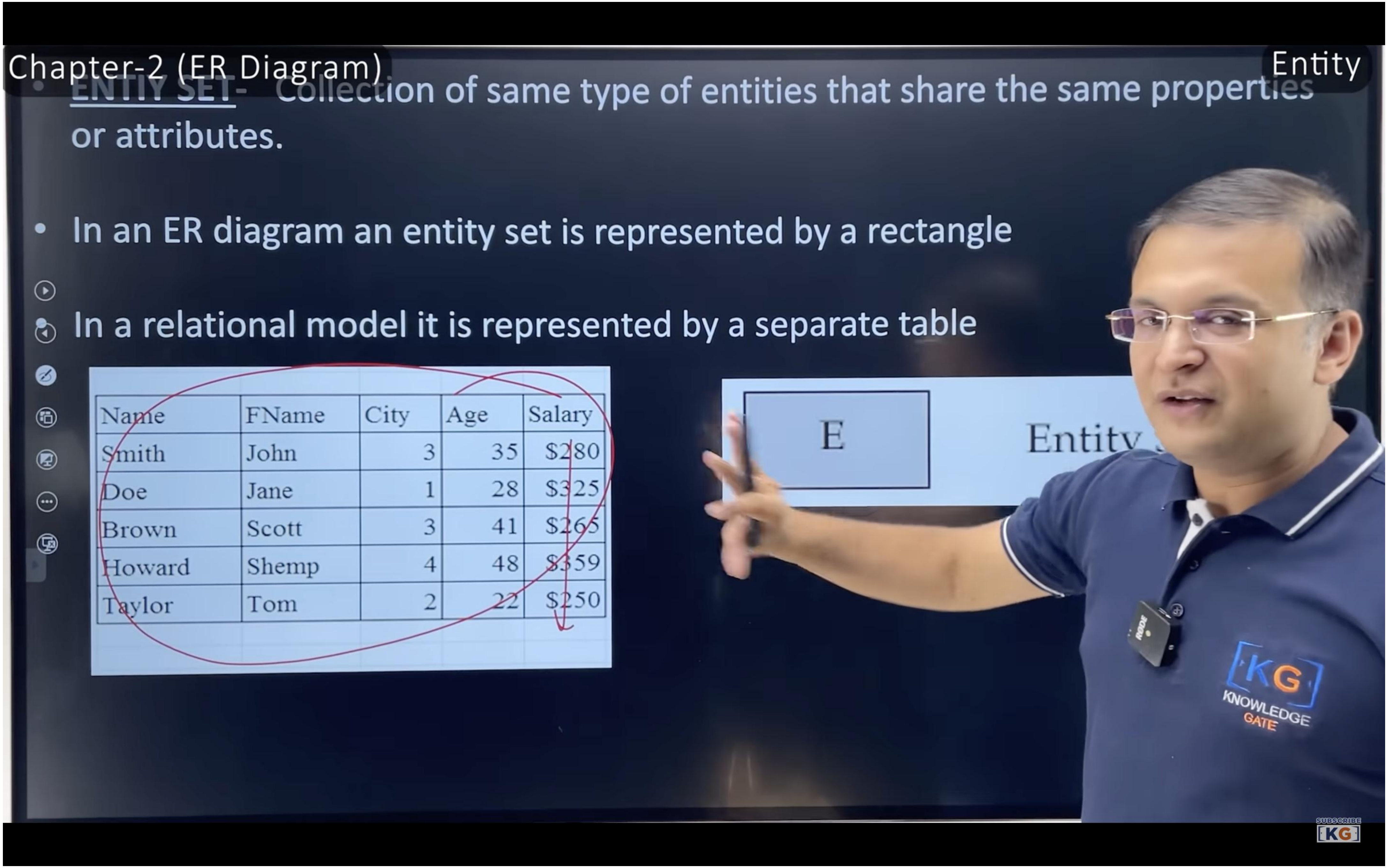
- An entity may be **concrete**, such as a person or a book, or it may be **abstract**, s a course offering, or a flight reservation.

Name	FName	City	Age	Salary
Smith	John	3	35	$280
Doe	Jane	1	28	$325
Brown	Scott	3	41	$265
Howard	Shemp	4	48	[illegible]
Taylor	Tom	2	22	[illegible]

- In ER diagram we cannot represent an entity, as entity is an instant not schema, and ER diagram is designed to understand schema
- In a relational model entity is represented by a row or a tuple or a rec table.



CustomerID	FirstName	LastName	Birthdate
XY001	John	Doe	April 18, 1929
BR092	Mary	Green	March 4, 1980
PD500	Francesca	de la Gillebert	September 12, 1959
WI308	John	Green	March 4, 1980



Chapter-2 (ER Diagram)
Entity
ENTITY SET- Collection of same type of entities that share the same properties or attributes.
In an ER diagram an entity set is represented by a rectangle
In a relational model it is represented by a separate table

Name	FName	City	Age	Salary
Smith	John	3	35	$280
Doe	Jane	1	28	$325
Brown	Scott	3	41	$265
Howard	Shemp	4	48	$359
Taylor	Tom	2	22	$250

E
Entity
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ATTRIBUTES

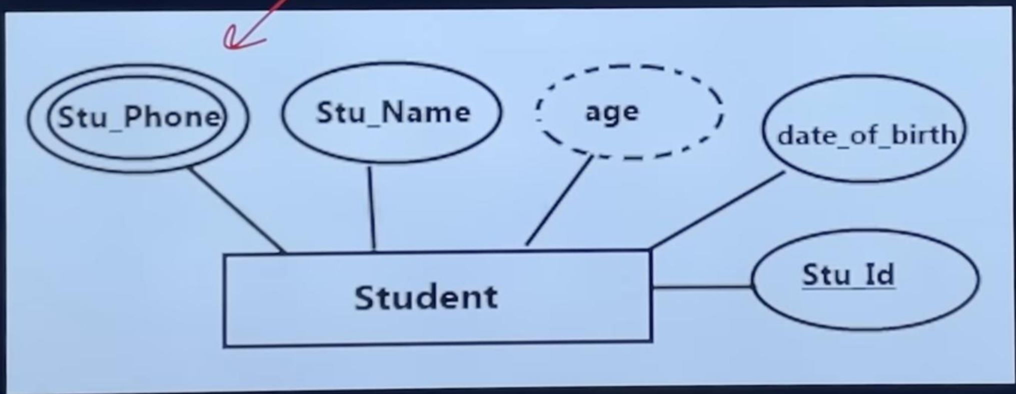
- Attributes are the units defines and describe properties and characteristics of entities.
- Attributes are the descriptive properties possessed by each member of an entity set. for each attribute there is a set of permitted values called domain.

Name	FName	City	Age	Salary
Smith	John	3	35	$280
Doe	[illegible]	1	28	$325
B[illegible]	[illegible]tt	3	41	$265
[illegible]d	Shemp	4	48	$359
[illegible]aylor	Tom	2	22	$250

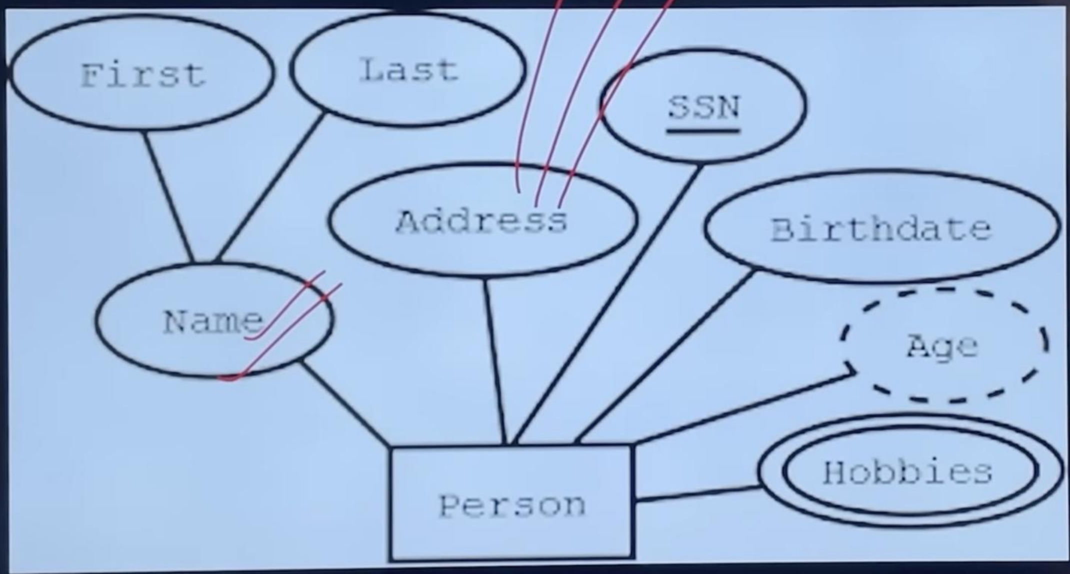
Types of Attributes

- **Single valued**- Attributes having single value at any instance of time for an entity. E.g. – Aadhar no, dob.

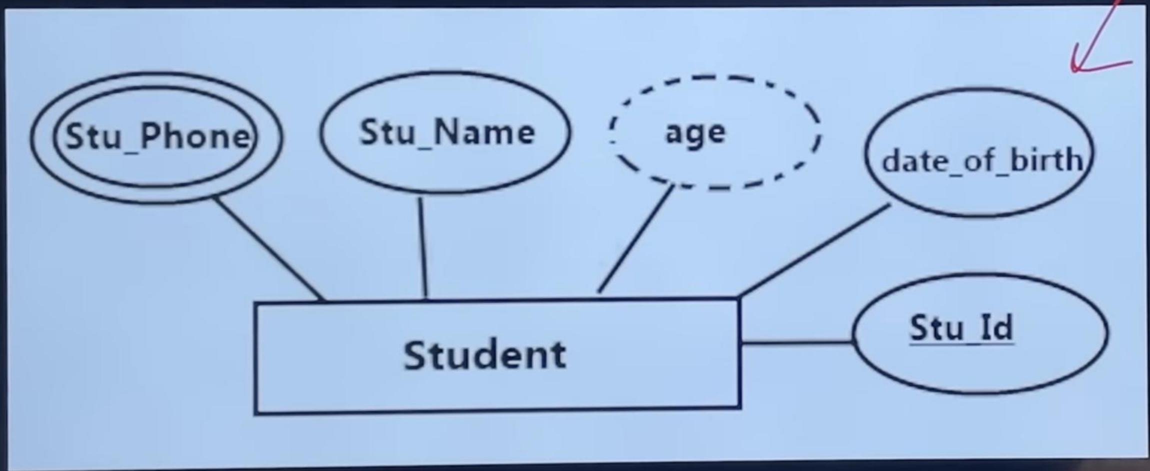
- **Multivalued** - Attributes which can have more than one value for an entity at same time. E.g. - Phone no, email, address.
  - A multivalued attribute is represented by a double ellipse in an ER diagram and by an independent table in a relational model.
  - Separate table for each multivalued attribute, by taking mva and pk of main table as fk in new table

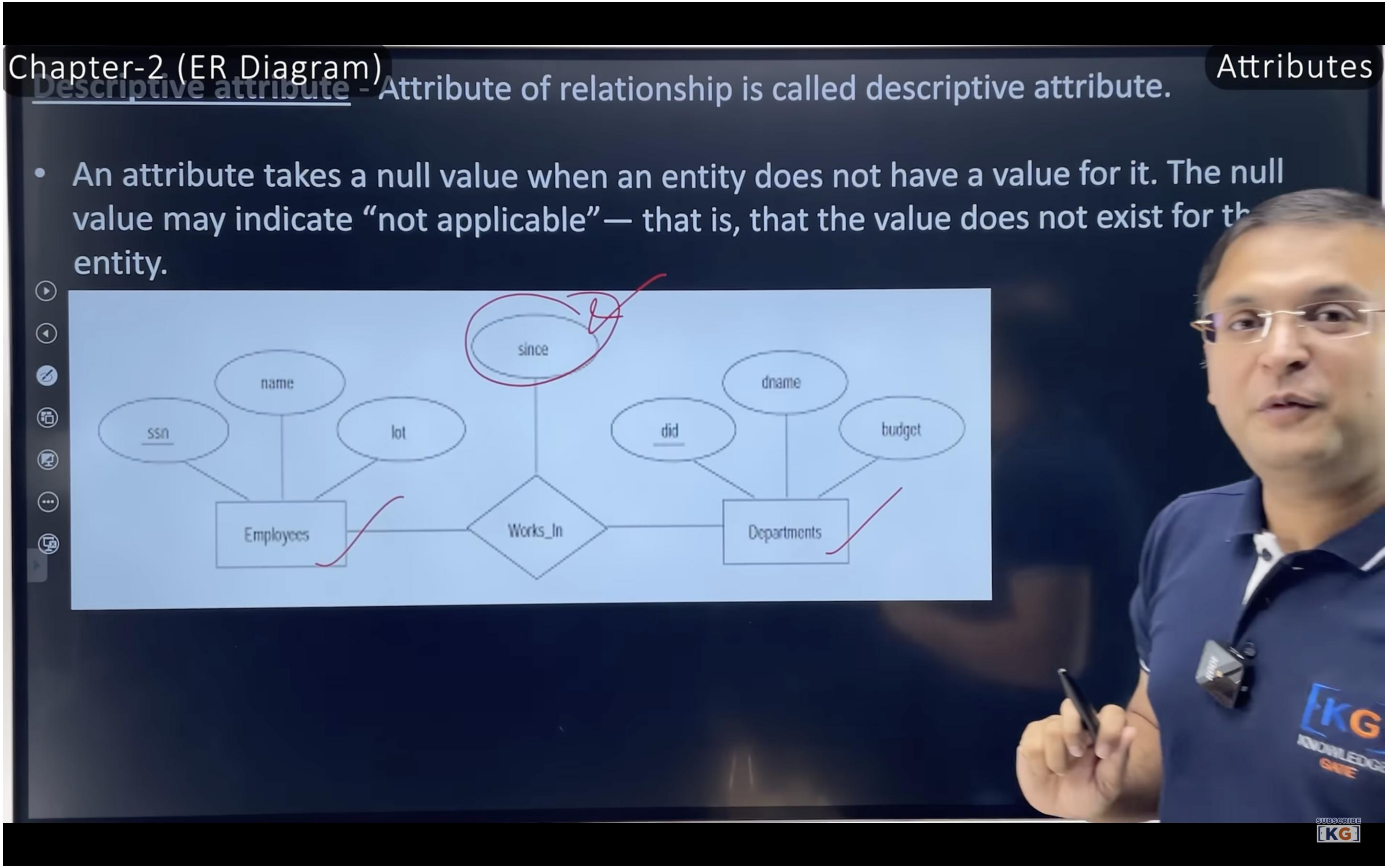


- **Simple** - Attributes which cannot be divided further into sub parts. E.g. Age
- **Composite** - Attributes which can be further divided into sub parts, as simple attributes. A composite attribute is represented by an ellipse co... an ellipse and in a relational model by a separate column.



- **Stored** - Main attributes whose value is permanently stored in database. E.g. date_of_birth
- **Derived** -The value of these types of attributes can be derived from values of other Attributes. E.g. - Age attribute can be derived from date_of_birth ar ttribute.




Chapter-2 (ER Diagram)
Attributes
Descriptive attribute - Attribute of relationship is called descriptive attribute.
• An attribute takes a null value when an entity does not have a value for it. The null value may indicate "not applicable"— that is, that the value does not exist for th
entity.
since
name
dname
ssn
lot
did
budget
Employees
Works_In
Departments
SUBSCRIBE
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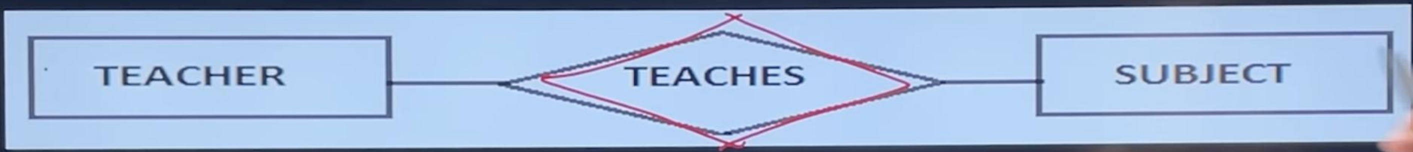
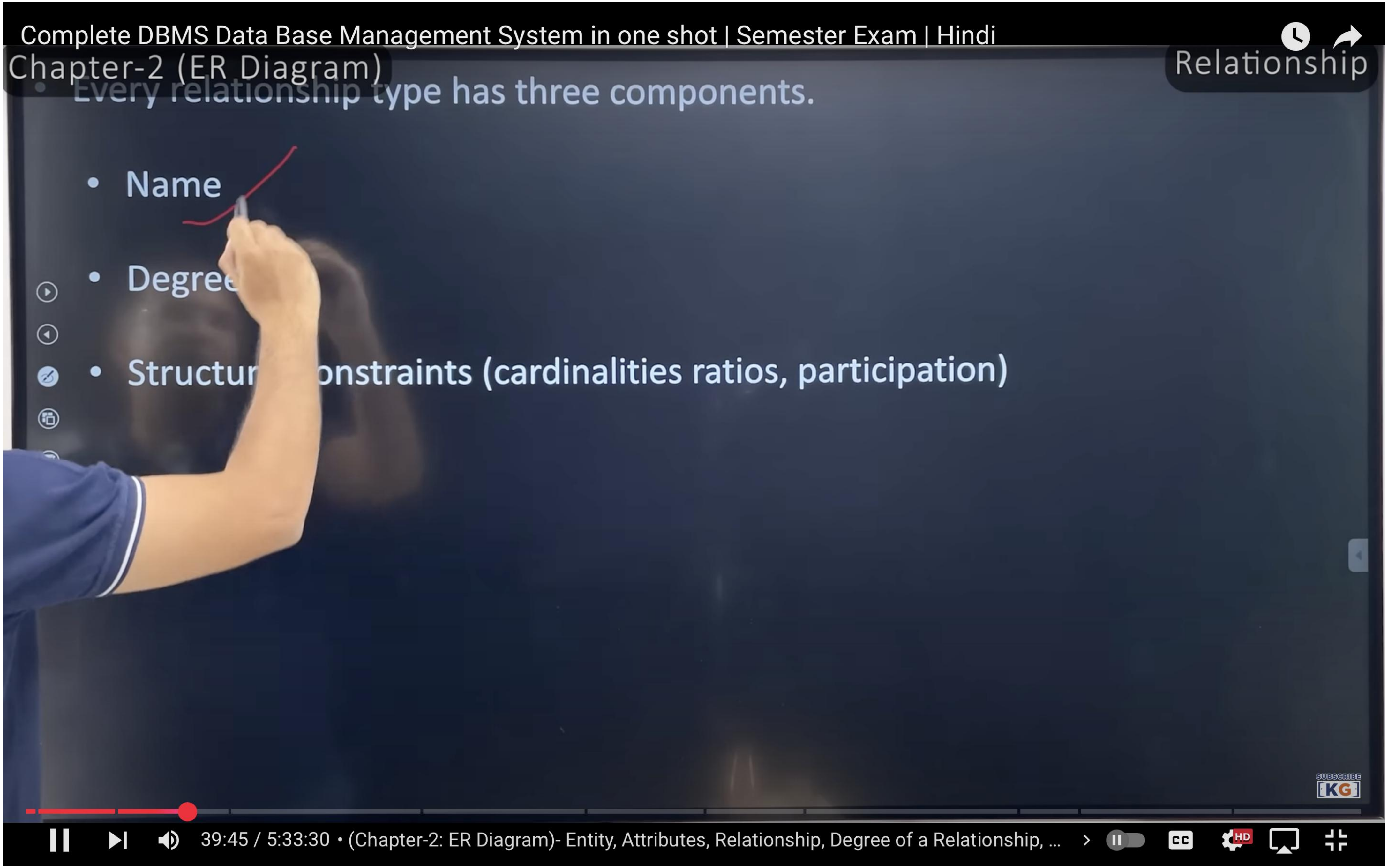
Relationship / Association

- Is an association between two or more entities of same or different entity set.
- In ER diagram we cannot represent individual relationship as it is an instance or data.





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Chapter-2 (ER Diagram)
Relationship
Every relationship type has three components.
Name
Degre
Structur
onstraints (cardinalities ratios, participation)
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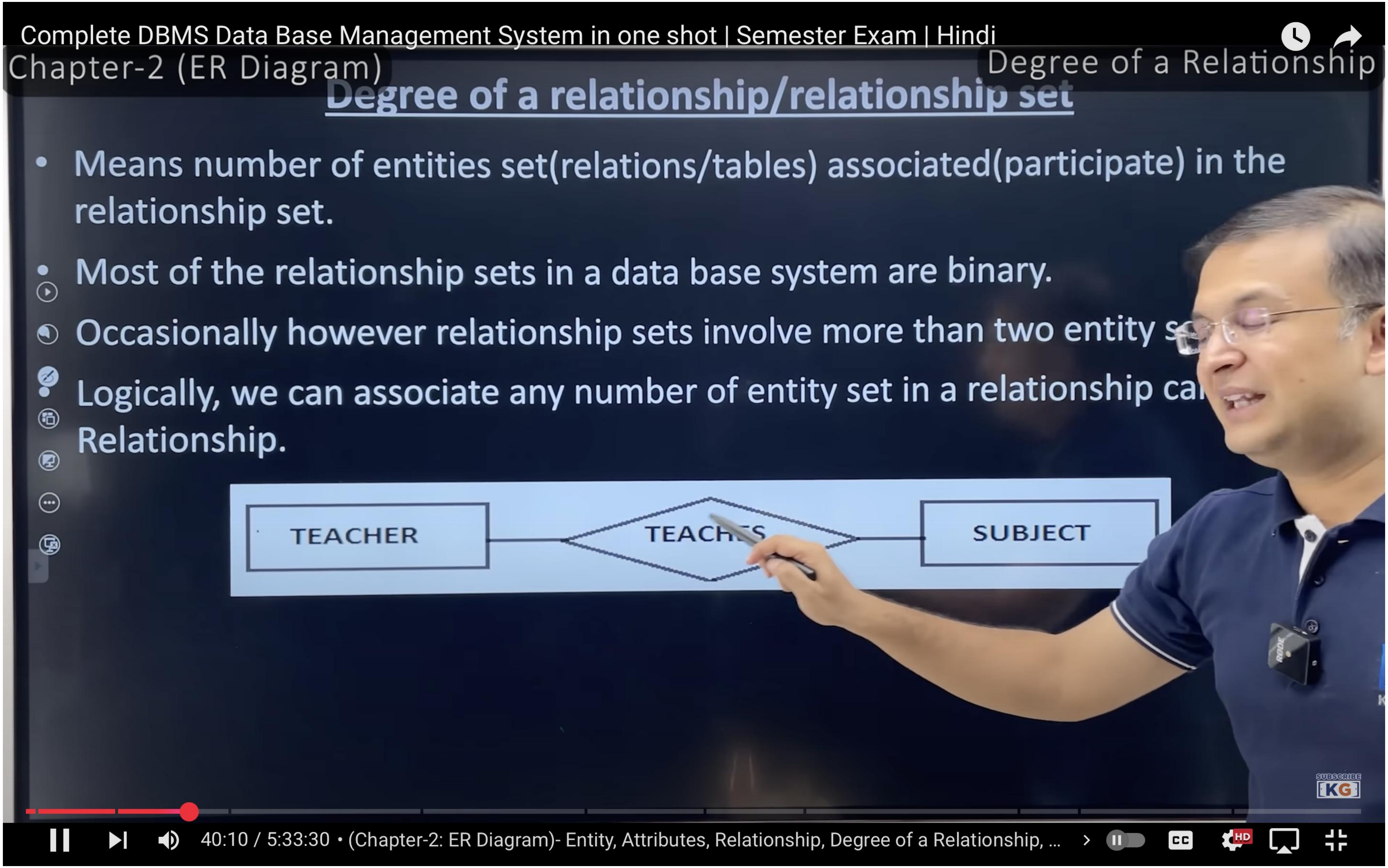
39:45 / 5:33:30 • (Chapter-2: ER Diagram)- Entity, Attributes, Relationship, Degree of a Relationship, ...
HD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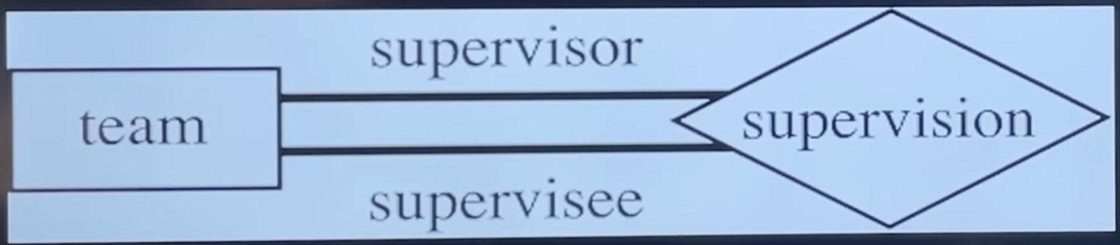
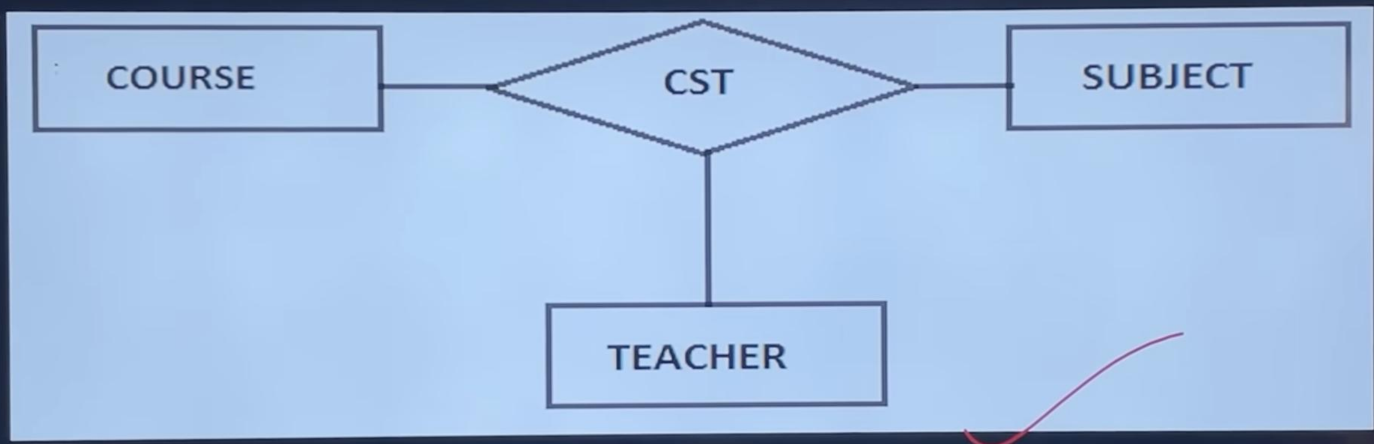
Chapter-2 (ER Diagram)
Degree of a Relationship
Degree of a relationship/relationship set
Means number of entities set(relations/tables) associated(participate) in the relationship set.
Most of the relationship sets in a data base system are binary.
Occasionally however relationship sets involve more than two entity s
Logically, we can associate any number of entity set in a relationship ca
Relationship.
TEACHER
SUBJECT
40:10 / 5:33:30 • (Chapter-2: ER Diagram)- Entity, Attributes, Relationship, Degree of a Relationship, ...

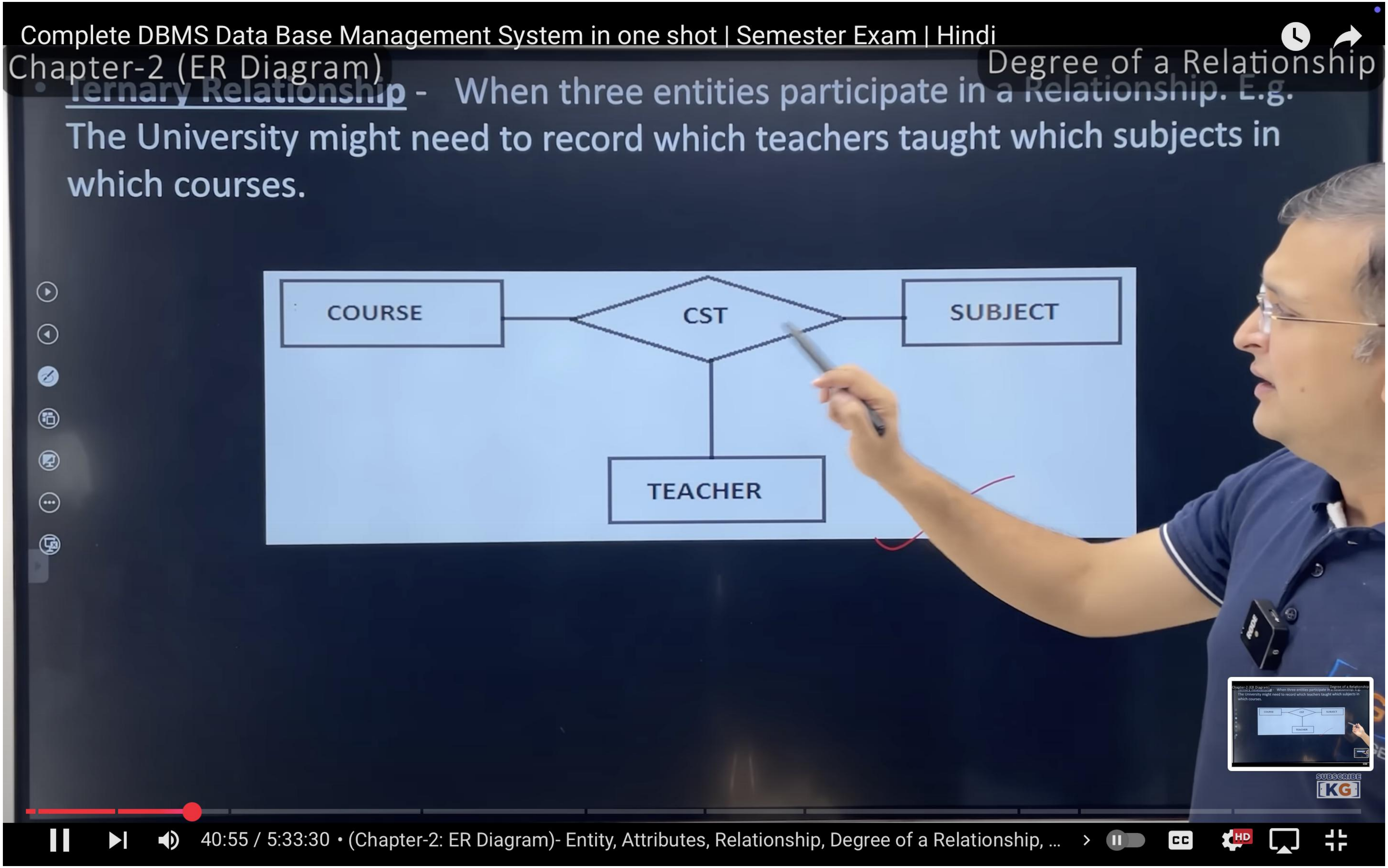
- **Unary Relationship** - One single entity set participate in a Relationship, means two entities of the same entity set are related to each other.
- These are also called as self -referential Relationship set.
- E.g.- A member in a team maybe supervisor of another member in team.



- **Binary Relationship** - Two entity sets participate in a Relationship. It is most common Relationship.

- **Ternary Relationship** - When three entities participate in a Relationship. E.g. The University might need to record which teachers taught which subjects in which courses.

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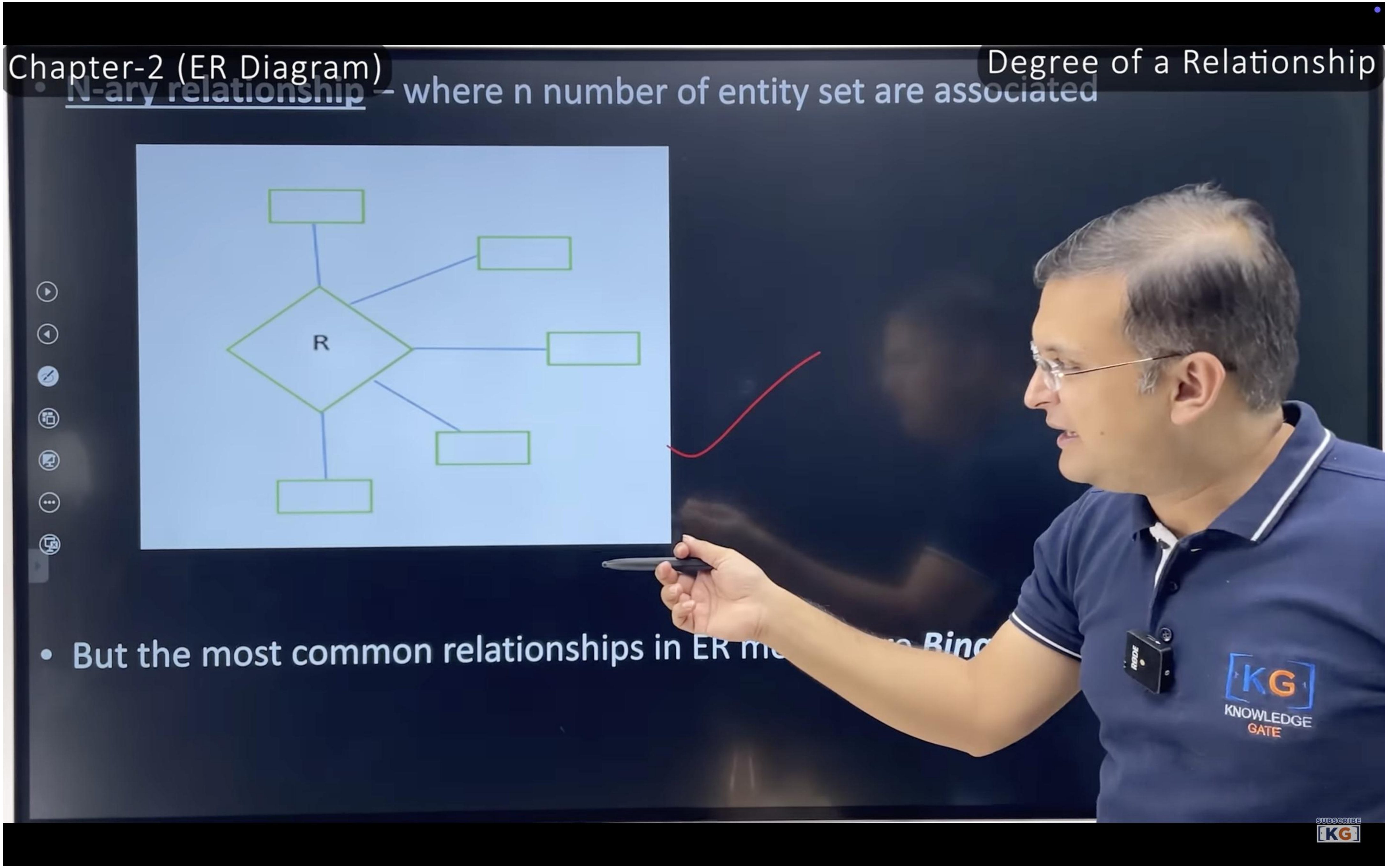
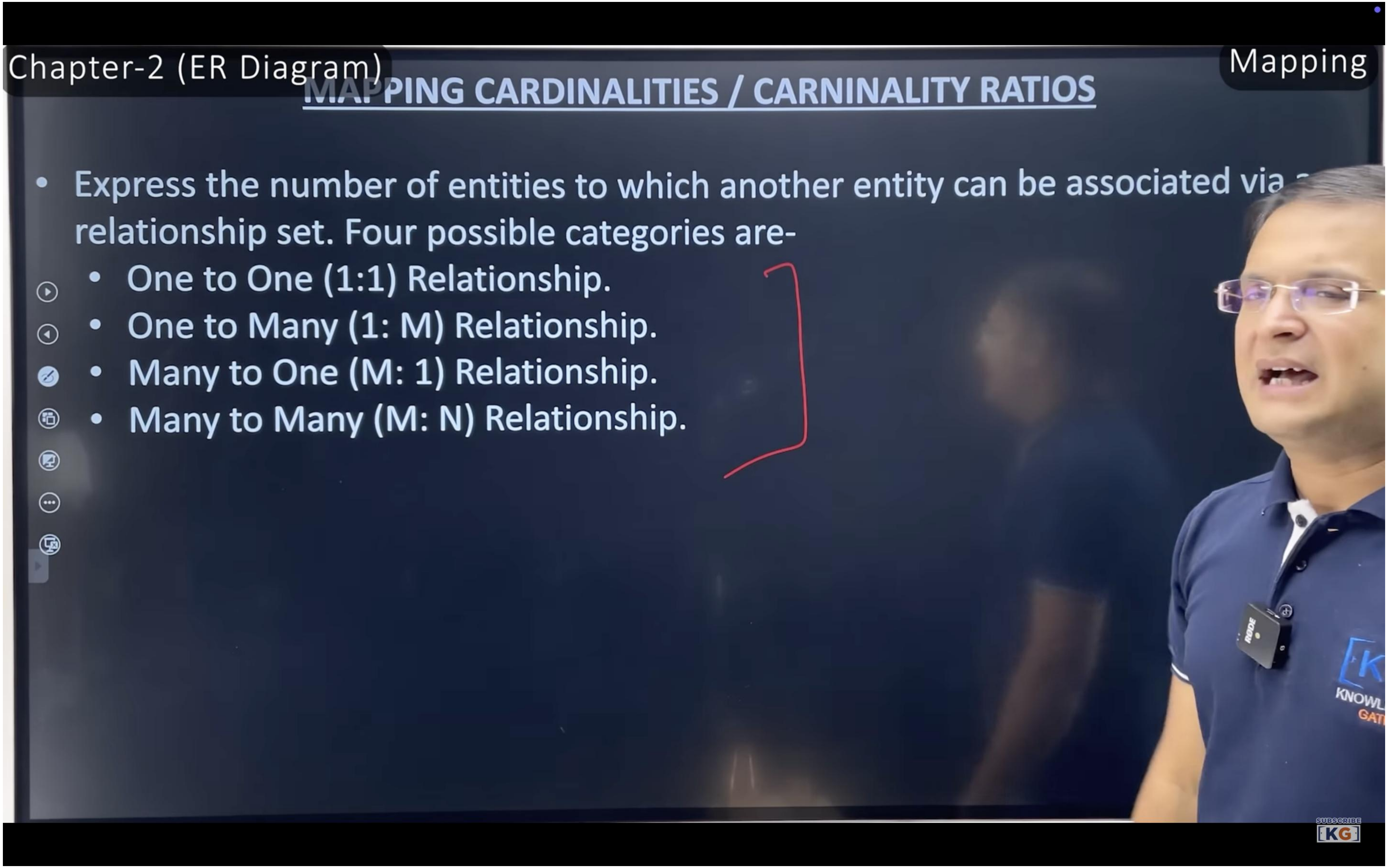
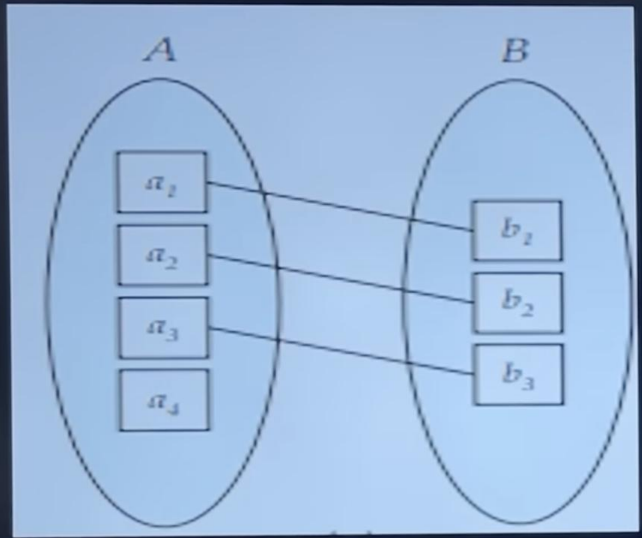
Chapter-2 (ER Diagram)
Degree of a Relationship
Ternary Relationship - When three entities participate in a Relationship. E.g. The University might need to record which teachers taught which subjects in which courses.
COURSE
CST
SUBJECT
TEACHER
40:55 / 5:33:30 • (Chapter-2: ER Diagram)- Entity, Attributes, Relationship, Degree of a Relationship, ...

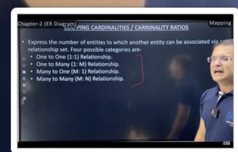

Chapter-2 (ER Diagram)
Degree of a Relationship
N-ary relationship – where n number of entity set are associated
R
But the most common relationships in ER
KG
KNOWLEDGE
GATE
SUBSCRIBE
KG


Chapter-2 (ER Diagram)
Mapping
MAPPING CARDINALITIES / CARNINALITY RATIOS
• Express the number of entities to which another entity can be associated via
relationship set. Four possible categories are-
• One to One (1:1) Relationship.
• One to Many (1: M) Relationship.
• Many to One (M: 1) Relationship.
• Many to Many (M: N) Relationship.

One to One (1:1) Relationship - An entity in A is associated with at most one entity in B, and an entity in B is associated with at most one entity in A.



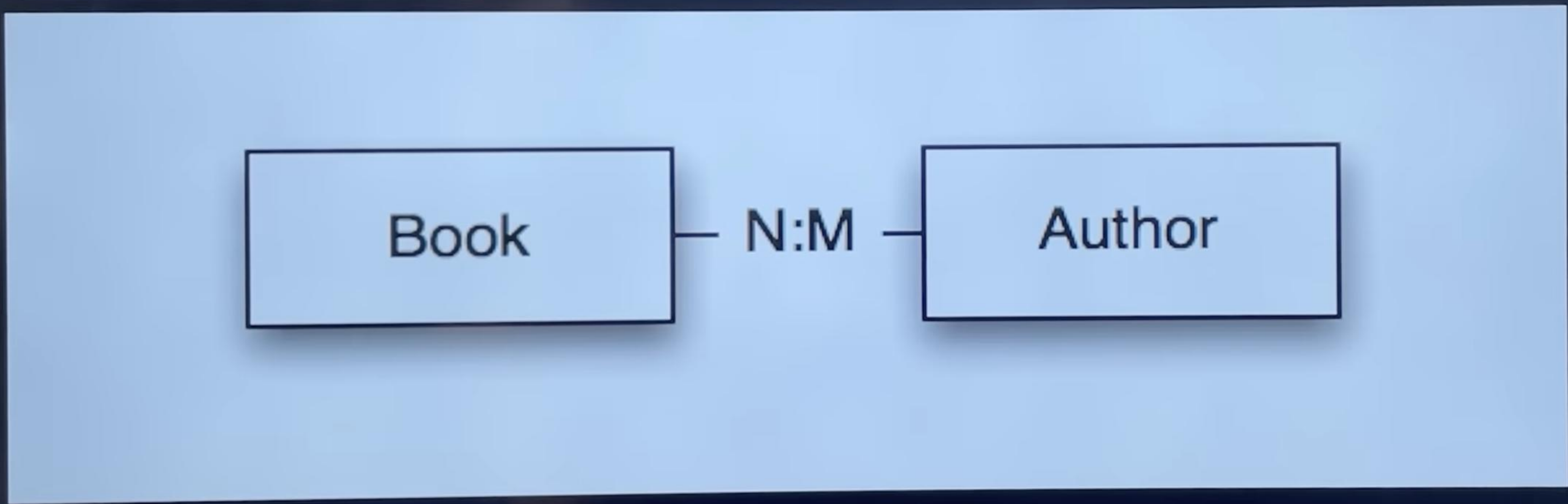
E.g.- The directed line from relationship set advisor to both entities set indicates th instructor may advise at most one student, and a student may have at most one advisor'.



- **PARTICIPATION CONSTRAINTS**- it defines participations of entities of an entity type in a relationship.

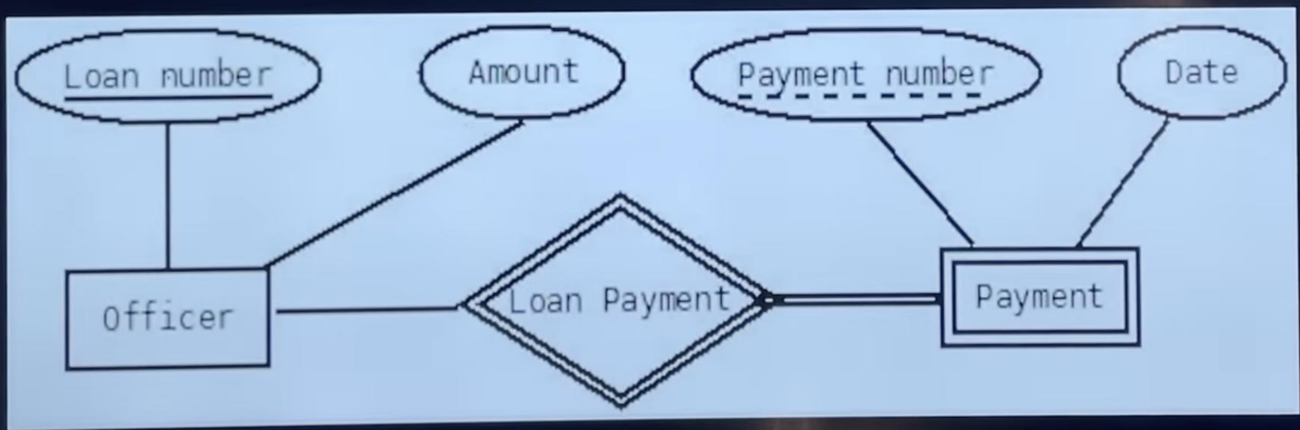
  - **Partial participation**

  - **Total Participation**



STRONG AND WEAK ENTITY SET

- An entity set is called strong entity set, if it has a primary key, all the tuples in the set are distinguishable by that key.
- An entity set that does not process sufficient attributes to form a primary key is cal weak entity set.
- It contains discriminator attributes (partial key) which contain partial informatio the entity set, but it is not sufficient enough to identify each tuple uniquely. Represented by double rectangle.



Chapter-2 (ER Diagram)

Conversion from ER Diagram to Relational Model

Conversion From ER Diagram To Relational Model

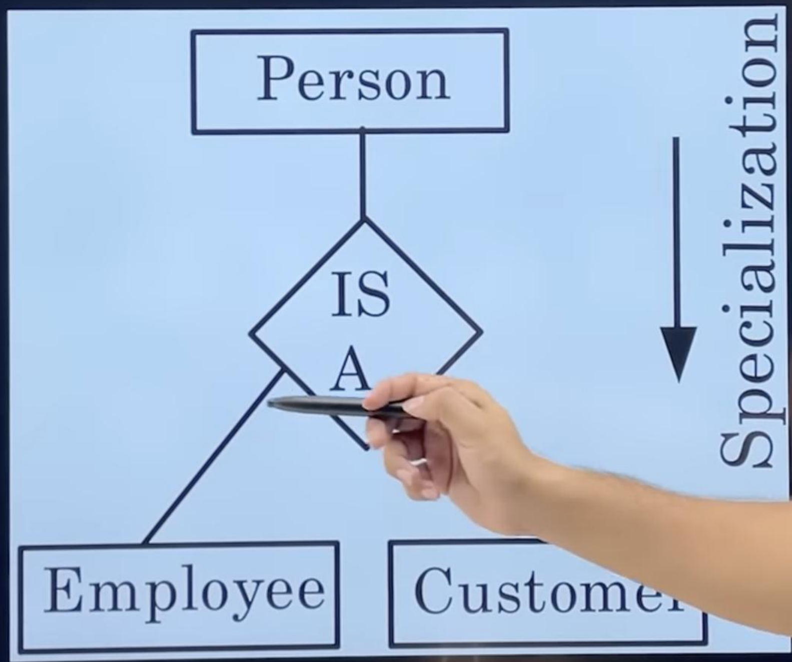
- Entity Set
  - Convert every strong, weak entity set into a separate table. In weak entity set we make it dependent onto one strong entity set **(identifying or owner entity set)**.
- Relationship
  - If Unary: No separate table is required, add a new column as fk which refer the pk of the same table.
  - if 1:1 No separate table is required, take pk of one side and put it as fk on other side, priority must be given to the side having total participation.
  - if 1:n or n:1 No separate table is required, modify n side by taking pk of 1 side a foreign key on n side.
  - If m-n Separate table is required take pk of both table and declare their combination as a pk of new table
  - (3 or More) Take the pk of all participating entity sets as fk and declare their combinations as pk in the new table.
- Attributes
  - Multivalued-A separate table must be taken for all multivalued attributes, where we take pk of the main table as fk and declare combination of fk and multivalued attribute are pk in the new table.
  - Composite Attributes-A separate column must be taken for all simple attributes of the composite attribute.

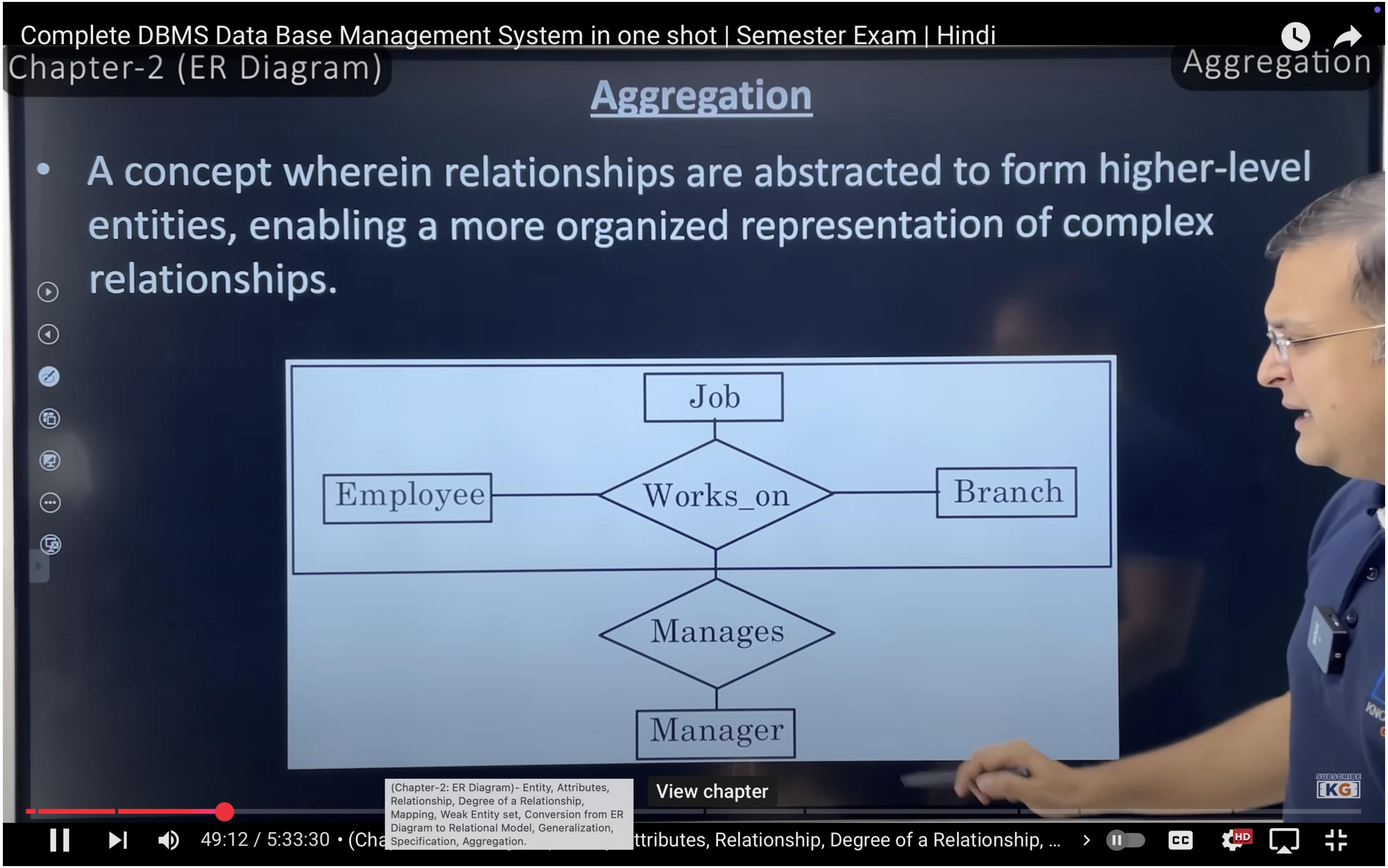
Generalization

- Involves merging two lower-level entities to create a higher-level entity.
- A bottom-up approach that builds complexity from simpler components.
- Highlights similarities among lower-level entity sets while hiding differe
- Leads to a simplified, structured data representation, aiding in databas and querying processes.

Specialization

- A process where a higher-level entity is broken down into more specific, lower-level entities.
- This top-down approach delineates complexity into simpler compon
- Acts as the converse of the generalization process, focusing on diffe properties rather than similarities.

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Chapter-2 (ER Diagram)
Aggregation
Aggregation
A concept wherein relationships are abstracted to form higher-level entities, enabling a more organized representation of complex relationships.
Job
Employee
Works_on
Branch
Manages
Manager
(Chapter-2: ER Diagram)- Entity, Attributes, Relationship, Degree of a Relationship, Mapping, Weak Entity set, Conversion from ER Diagram to Relational Model, Generalization, Specification, Aggregation.
View chapter
49:12 / 5:33:30 • (Cha
ttributes, Relationship, Degree of a Relationship, ...
KG

ADVANTAGES OF E-R DIGRAM

- Constructs used in the ER diagram can easily be transformed into relational tables.
- It is simple and easy to understand with minimum training.

DISADVANTAGE OF E-R DIGRAM

- Loss of information content
- Limited constraints representation
- It is overly complex for small projects